DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISAL

FINAL EXAMINATION-2022
Course Title: Theory of Computation
Course Code: CSE-3203
3rd Year, 2nd Semester, Session: 2016-17

Time: 3 hours **Marks:** 60

Answer any Five (5) Questions from the followings.

1. a) Analyze the terms: Theorems, Lemmas and Corollaries. 4
   b) What is Automata Theory? Describe the applications of Finite Automata. 4
   c) A containment hierarchy of classes of formal languages has been classified as The Chomsky Hierarchy. Describe with suitable figure(s). 4

2. a) Define push down automata with example. 3
   b) Give pushdown automata that recognize the following languages: 5
      (a) $A = \{ w \in \{0, 1\}^* \mid w \text{ contains at least three 1s} \}$
      (b) $B = \{ w \in \{0, 1\}^* \mid w = w^R \text{ and the length of w is odd} \}$
   c) Use the pumping lemma to prove that the language $A = \{ 0^{2n} 1^{3n} 0^n \mid n \geq 0 \}$ is not context free. 4

3. a) What is Turing Machine? Give advantages of it. 3
   b) Construct a Turing Machine that accepts the language of palindromes over {a,b} . Also specify the moves to trace the strings abaa,abba,aabaa. 6
   c) What is meant by a halting problem in a Turing Machine? Explain with an example. 3

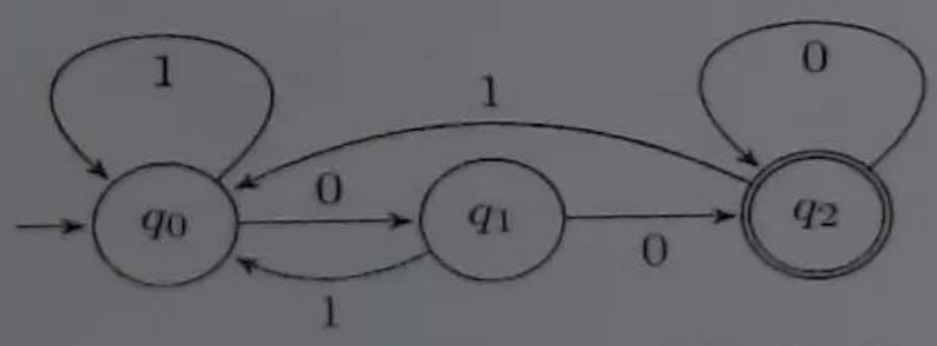
4. a) Let, $\Sigma = \{0, 1\}$
   Construct a DFA for the following language:
   $L = \{w \mid w \text{ is a binary string that has even number of 1s and even number of 0s}\}$ 6
   b) Construct a NFA for the following: Strings where the first symbol is present somewhere later on at least once. 6

5. a) Build an NFA for the following language: $L = \{ w \mid w \text{ ends in } 01\}$ 4
   b) Describe the Advantages & Caveats for NFA. 3
   c) Convert the NFA from Question 5(a), to DFA. 5

6. a) Why explicit ε-transitions in finite automata is important? 2
   b) Build an ε-NFA for the following language: 4
   $L = \{w \mid w \text{ is empty, or if non-empty will end in } 01\}$
   c) Convert ε-NFA to DFA based on Question 6(b) 6

7. a) Describe the relation between Regular Expressions (RE) and Finite Automata. Show with figure that they are interchangeable. 6
   b) Convert the following RE to ε-NFA: (0+1)*01(0+1)* 6

8. a) Construct a context-free grammar for the following DFA: 6



   b) Show that the grammar ({S}, {a, b}, R, S) with rules $R = S \rightarrow aS \mid aSbS \mid \epsilon$ is ambiguous. 4
   c) Does a push down automata have memory? Justify. 2

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